

Mini Project Report

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
College of Engineering and Technology
Department of Electronics and Communication Engineering
<i>21ECC211L / Devices and Digital IC Lab</i>
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Lab code & Title : 21ECC211L & Devices and Digital IC Lab

Year & Semester : 2nd year & 3rd semester

Project Title : 0-99 Digital counter using CD4017 IC

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Mark split up ↓	BARATH SHAGAR M	NARAYANA SWAMY MR	VARUN PR	SUGANDHAN MSA
Novelty of the project (1 mark)				
Level of understanding of the Project (2 marks)				
Contribution of the Project Work (1 mark)				
Project Report (1 Mark)				
Total (5 Marks)				

REPORT VERIFICATION

Date:

Signature of Lab Supervisor

Mini Project Title	0-99 Digital counter using CD4017 IC
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OBJECTIVE

The objective of this project is to design and implement a digital counter system that counts manually from 0 to 99 using two CD4017 decade counter ICs. The system is controlled by a pushbutton to generate clock pulses manually, allowing the user to increment the count with each button press. This project aims to demonstrate the functionality of a simple digital counting circuit, focusing on manual control and proper debouncing techniques to ensure reliable operation. The system will visually display the count using LEDs, and will reset automatically after reaching the value of 99.

ABSTRACT:-

This project presents the design and implementation of a digital counter capable of counting from 0 to 99 using two CD4017 decade counter ICs. The counting mechanism is controlled manually through a pushbutton that generates clock pulses to increment the count. LEDs are used to display the current count visually, with the system resetting to 00 after reaching 99. Debouncing techniques are employed to ensure accurate counting and prevent errors caused by mechanical bouncing of the pushbutton. The project demonstrates the practical application of digital counters in simple electronics systems, offering manual control over the counting process.

INTRODUCTION:-

Digital counters are essential components in a wide range of electronic systems, from simple displays to complex computing devices. They are used to count events, time intervals, or any sequence requiring a numerical representation. In this project, a manual digital counter is implemented using two CD4017 decade counter ICs to count from 0 to 99. The counting is controlled by a pushbutton that generates clock pulses, incrementing the count with each press.

The CD4017 is a widely used decade counter that can count from 0 to 9, making it suitable for this project's objective of displaying two-digit numbers. By cascading two CD4017 ICs, we can achieve a counting range from 0 to 99, with one IC controlling the units place and the other the tens place. LEDs are connected to the outputs of each IC to visually display the current count. Additionally, a reset mechanism is implemented to return the count to 00 after reaching 99.

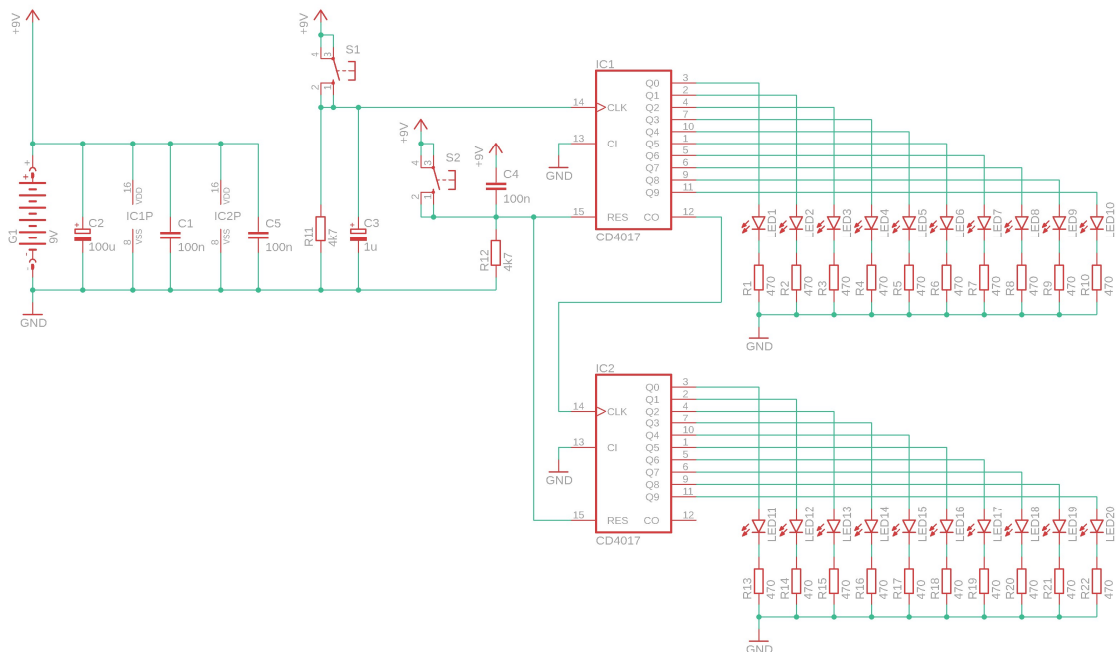
To ensure reliable counting, debouncing techniques are incorporated to eliminate the effects of pushbutton bouncing, which can cause multiple unwanted counts. This project provides a practical demonstration of manual digital counting, offering insights into the design and operation of basic digital circuits used in various applications.

SOFTWARE&HARDWARE REQUIREMENT/DESCRIPTION:-

- 2 × 830-pin breadboard
- 1 × 9V block battery + battery clip
- 2 × CD4017 counter IC
- 2 × pushbuttons
- 20 × LED of your choice
- 20 × 470Ω resistor
- 2 × 4.7kΩ resistor
- 1 × 1μF electrolytic capacitor
- 1 × 100μF electrolytic capacitor
- 3 × 100nF ceramic capacitor

And some connecting wires.

CIRCUIT DIAGRAM AND DISCRPTION:



CIRCUIT DISCRIPTION:

Power Supply Section:

- *G1*: A 9V battery is used to power the circuit. The positive terminal is connected to various components, while the negative terminal (GND) serves as the common ground.
- *Capacitors (C1, C2, C5)*: These are decoupling capacitors used to filter any noise in the power supply, providing stable voltage to the ICs.
 - *C1 and C5*: 100nF capacitors.
 - *C2*: A 100 μ F capacitor, providing additional filtering.

Clock Generation Section:

- *S1 (Switch 1)*: This acts as a manual clock input. Every time the button is pressed, it sends a pulse to increment the counter.
- *R1 and C3*: These components form a debouncing circuit that cleans the clock signal from the switch (S1), ensuring that only one pulse is registered per button press.
 - *R1*: A 4.7k Ω resistor.
 - *C3*: A 1 μ F capacitor.

Reset Section:

- *S2 (Switch 2)*: This switch resets the counter to zero when pressed.
- *R2 and C4*: Another debouncing circuit to prevent bouncing when the reset button (S2) is pressed.
 - *R2*: A 4.7k Ω resistor.
 - *C4*: A 100nF capacitor.

Counting Section (CD4017 IC):

The circuit includes *two CD4017 ICs*:

- *IC1 and IC2 (CD4017 ICs)*: These are decade counters with 10 outputs. Each output goes high in sequence as the clock signal advances.
 - *Pin 14 (CLK)*: The clock input. It advances the counter on each rising edge.
 - *Pin 15 (RESET)*: Resets the counter when activated.
 - *Pin 13 (CI - Carry In)*: Connected to GND in both ICs.
 - *Pin 12 (CO - Carry Out)*: The carry-out of the first IC is connected to the clock input of the second IC (IC2) to allow the second CD4017 to count after the first completes its cycle.

Output Section (LED Display):

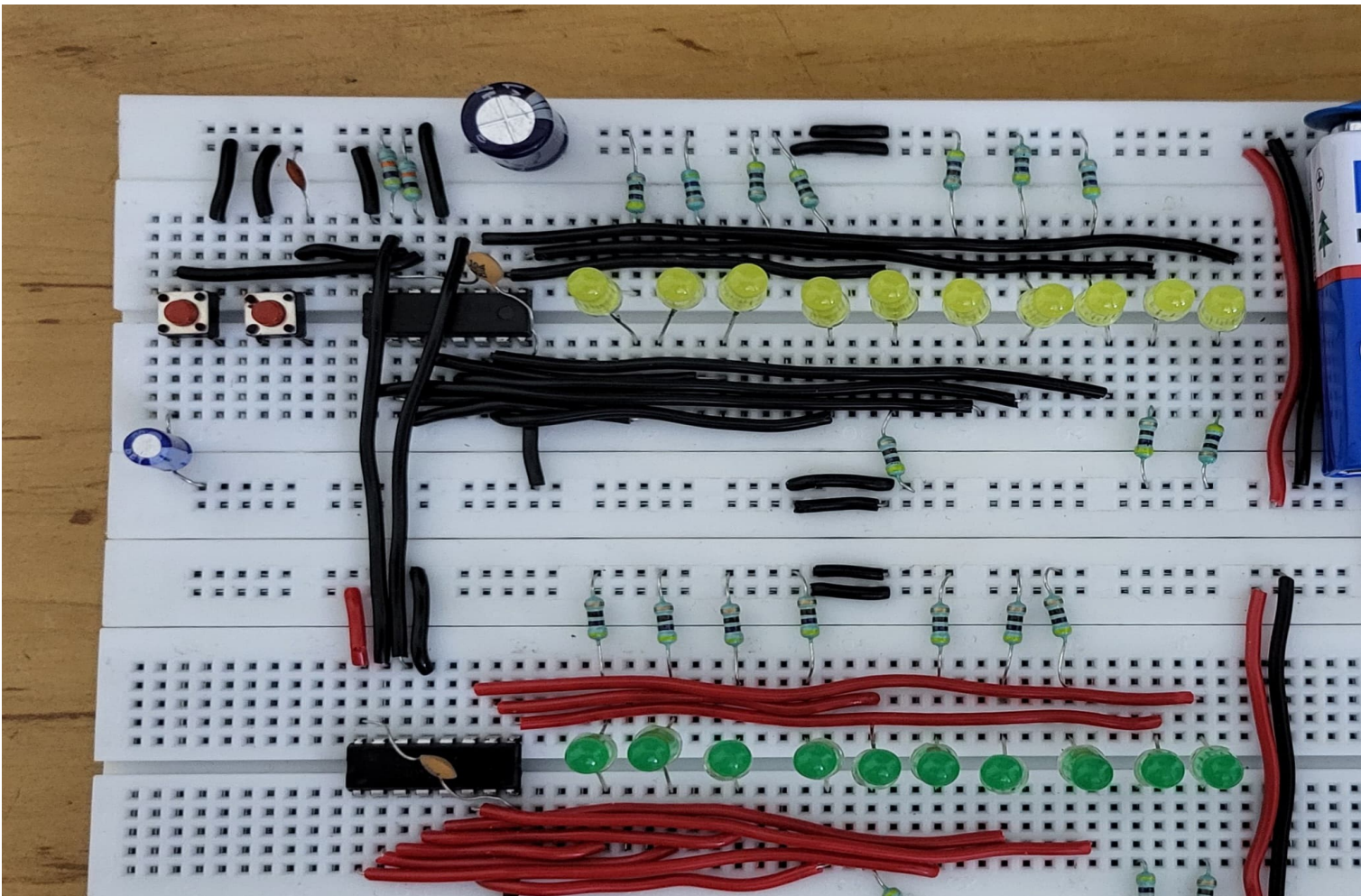
- Each *LED (LED1 - LED20)* represents the output of the CD4017 ICs.
- *Resistors (R1 - R22)*: Each LED is connected through a 470 Ω resistor to limit current and prevent burning out the LEDs.

Function:

- The circuit counts manually from 0 to 99 using the *two CD4017 ICs*.
- Pressing *S1* increments the count, lighting up the corresponding LED.
- Pressing *S2* resets the count back to 0.
- The first CD4017 counts from 0 to 9, then the second CD4017 takes over and counts from 10 to 19, and so on, lighting up each corresponding LED.

RESULTS AND DISCUSSION:-

The manual digital counter system was successfully implemented using two CD4017 decade counter ICs, a pushbutton for manual clock pulses, and LEDs to display the count from 0 to 99. The counter operated as expected, with each button press incrementing the count by one, and the LEDs correctly reflecting the current value of the units and tens digits. Upon reaching 99, the system reset to 00, demonstrating the correct functionality of the reset mechanism.



CONCLUSION:-

This project successfully designed and implemented a manual digital counter system using two CD4017 decade counter ICs. The system is capable of counting from 0 to 99, with each count increment triggered by a pushbutton. The LEDs provided a clear visual representation of the current count, and the system automatically reset after reaching 99. Proper debouncing techniques were employed to ensure reliable operation, preventing errors from pushbutton bounce.

The project effectively demonstrated the practical application of basic digital counters in electronic systems. It also highlighted the simplicity and flexibility of using CD4017 ICs for counting tasks. With potential for further enhancements, such as an LCD display or memory storage, this project lays a solid foundation for understanding and implementing manual counting circuits in various real-world application.

REFERENCES:

https://www.youtube.com/redirect?event=video_description&redir_token=OUFFLUhqbWVDX2FHyl1vUEglV21NcEhoWjBYUFZhVUhPZ3xBQ3Jtc0trODA0SGZZUmJEWXVKZ2w1WlZ1enJqQ2pUNTFhYTlIZzQwUfNjNGFjekdKRk5yeTBYQlNsZFJYdUpfS3dyQzhNUWhINThtTTRwRjZrNURBRHdncS0xWXNIR3lDRksZVzM4QmthaWotQmxzdnMzd2M3WQ&q=http%3A%2F%2Fwww.friendlywire.com%2Ftutorials%2Fcd4017&v=BZjO2TuS2Vk

https://youtu.be/BZjO2TuS2Vk?si=2MPAZ4L_C4vIVSIG

